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Windows PowerShell
PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W7D2_Haystack_Retrieval> python w7d2_haystack_retrieval.py
=====
W7D2 - HAYSTACK RETRIEVAL: BM25 & DENSE
=====

Found PDF files:
- artificial_intelligence.pdf
- cloud_computing.pdf
- Document 1.pdf
- software_design.pdf
- software_engineering.pdf

Total PDF files found: 5

=====
STEP 2 - PDF CONVERSION
=====
Converted documents: 5

=====
STEP 3 - DOCUMENT CLEANING
=====
Clean documents: 5

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STEP 4 - DOCUMENT CHUNKING
=====
Total document chunks: 2639

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STEP 5 - BM25 DOCUMENT STORE
=====
BM25 document store contains: 2639 documents
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Windows PowerShell
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STEP 6 - BM25 RETRIEVER
=====
BM25 retriever created successfully.
BM25 Haystack pipeline created successfully.

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BM25 RETRIEVAL RESULTS - 10 QUESTIONS
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Question 1: What is artificial intelligence?
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Result 1
Source: artificial_intelligence.pdf
BM25 Score: 19.8468
Content: We also examine some of the ethical debates surrounding this topic, debates that are and will consider to be at the forefron
t of the conversations surrounding AI in the near future. We then go into a discussion on the technique behind most modern AI systems
: Machine Learning. We cover some of the ba...

Result 2
Source: artificial_intelligence.pdf
BM25 Score: 19.3626
Content: - My favorite definition is that AI is the ability of a computer to do tasks that generally require human intelligence. - An
other definition is a computer system that is able to do something that it was not explicitly programmed to do - In general, most def
initions are centered around the idea compu...

Result 3
Source: artificial_intelligence.pdf
BM25 Score: 18.3087
Content: Lecture #25: Artificial Intelligence and Machine Learning CS106E Spring 2018, Payette & Lu In this lecture, we study Artific
ial Intelligence and Machine Learning. We start by defining and looking at the history of Artificial Intelligence. We explore the tec
hnological advances that allowed the recent...

Question 2: What is machine learning?
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Windows PowerShell

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Result 1
Source: artificial_intelligence.pdf
BM25 Score: 17.1469
Content: We also examine some of the ethical debates surrounding this topic, debates that are and will consider to be at the forefront of the conversations surrounding AI in the near future. We then go into a discussion on the technique behind most modern AI systems : Machine Learning. We cover some of the ba...

Result 2
Source: artificial_intelligence.pdf
BM25 Score: 16.4070
Content: determining what's in a picture), prediction (i.e. which Netflix shows is this user most likely to watch), decision making (i.e. should this autonomous car turn), or data generation (i.e. human speech synthesis). Machine learning data most frequently takes the form of input-label pairs ( x, y ) ...

Result 3
Source: artificial_intelligence.pdf
BM25 Score: 16.3927
Content: - Do we allow systems to update themselves? - Does the programmer have a responsibility to limit the intelligence of his/her system?
AI Frontiers/Current Work/Future Work - Robotics - Natural Movement - Environment Understanding - Natural Language Understanding/Processing - Bias Detection - Summary ...

Question 3: What is cloud computing?
-----
Result 1
Source: cloud_computing.pdf
BM25 Score: 15.8607
Content: This definition attempts to encompass all of the various cloud approaches. Definition of Cloud Computing: Cloud computing is a model for enabling convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) t...

Result 2
Source: cloud_computing.pdf
BM25 Score: 15.3341
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Windows PowerShell

Content: The NIST Definition of Cloud Computing Authors: Peter Mell and Tim Grance Version 15, 10-7-09 National Institute of Standards and Technology, Information Technology Laboratory Note 1: Cloud computing is still an evolving paradigm. Its definitions, use cases, underlying technologies, issues, risks, a...

Result 3
Source: software_engineering.pdf
BM25 Score: 14.4143
Content: Designing an efficient distributed computing system is relatively difficult. Using multiple processors or cores could be expensive. Dependency of one process on another process could introduce latency. Example Applications Telephone and cellular networks, internet, World Wide Web networks, distribut...

Question 4: What is deep learning?
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Result 1
Source: software_engineering.pdf
BM25 Score: 14.8890
Content: Machine learning (ML) enables computer systems to learn from experiences and to use the knowledge gained to make smart decisions - to become artificially intelligent. Deep learning uses artificial neural network models for learning and making predictions. Everyone expects all systems they use to ...

Result 2
Source: software_engineering.pdf
BM25 Score: 14.6047
Content: Process Assessment and Improvement in Agile [9*,c11][28] Agile methods (e.g., the scrum project management method) introduce what they call retrospectives at the end of each iteration. The objective of the retrospective is to analyze what went well and what did not go well, to understand why, and...

Result 3
Source: software_engineering.pdf
BM25 Score: 14.3997
Content: Seacord, The CERT C Secure Coding Standard, Addison-Wesley Professional, 2016. [13] R. Fagin, "A Normal Form for Relational Databases that is based on Domains and Keys," ACM Transactions on Database Systems, Vol. 6, No. 3, ACM, September 1981 [14] I. Goodfellow, Y. Bengio, A. Courville, Deep Learning...

Question 5: What is natural language processing?
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Windows PowerShell

Result 1
Source: software_engineering.pdf
BM25 Score: 21.0051
Content: 9.5. Natural Language Processing Natural language processing (NLP) is a crucial part of AI systems, enabling users to interact with the AI systems in a way that is similar to how they interact with other humans. AI systems understand human languages and execute commands delivered in those languages...

Result 2
Source: artificial_intelligence.pdf
BM25 Score: 19.1318
Content: - Do we allow systems to update themselves? - Does the programmer have a responsibility to limit the intelligence of his/her system?
AI Frontiers/Current Work/Future Work - Robotics - Natural Movement - Environment Understanding - Natural Language Understanding/Processing - Bias Detection - Summary ...

Result 3
Source: software_engineering.pdf
BM25 Score: 19.0512
Content: ISO/IEC/IEEE 29148 [26], as well as [1*, c10-14], [5, c4], [6, c16], and many others offer templates for requirements documentation. 4.1. Unstructured Natural Language Requirements Specification [1*, c11] [2*, s4.4.1] Natural language requirements specifications express requirements in common, or...

Question 6: What are neural networks?
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Result 1
Source: software_engineering.pdf
BM25 Score: 19.1408
Content: This model is commonly used in health care and banking applications. Logistic Regression model is a statistical model primarily used for classifying dependent variables from given independent variables. Artificial Neural Networks are inspired by biological neural networks in a brain. The systems ...

Result 2
Source: software_engineering.pdf
BM25 Score: 18.6892
Content: Others try to characterize how these dynamic variables evolve over time. The study of neural dynamics can be further subd
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Windows PowerShell

Result 3
Source: artificial_intelligence.pdf
BM25 Score: 18.3858
Content: Choosing a model involves considering a number of trade-offs including running time, amount of data required, and performance of the model. In addition, models often make assumptions about the underlying structure of the data, which can impact the performance of the model if the assumptions are not ...

Question 7: What is data processing?
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Result 1
Source: software_engineering.pdf
BM25 Score: 13.7116
Content: • Single instruction, multiple data stream (SIMD) architecture. • Multiple instruction, single data stream (MISD) architecture. • Multiple instruction, multiple data stream (MIMD) architecture. Variants of these architectures include array processing, parallel processing, and associative processing...

Result 2
Source: software_engineering.pdf
BM25 Score: 13.6348
Content: The design and construction processes should accommodate string and character set issues, including which character set is used, what kinds of strings are used, how to maintain the strings without changing the code and how to translate the strings into different languages with minimal impact on the ...

Result 3
Source: software_engineering.pdf
BM25 Score: 11.7662
Content: This content area discusses various computer architectures and organizations a system or software architect needs to know. 2.1. Computer Architecture [8*, C1.1] Architecture describes what the computer or system does, and its components, such as memory, data storage devices, graphics, and the computer...

Question 8: What is software architecture?
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Result 1
Source: software_engineering.pdf
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Windows PowerShell
Source: software_engineering.pdf
BM25 Score: 10.7777
Content: Much like building architecture, software architecture is outward-looking; it considers a system's context beyond its boundaries including the people, organizations, software, hardware and other devices with which the system must interact. Software Architecture /The Senses of "architecture..."

Result 2
Source: software_engineering.pdf
BM25 Score: 10.5254
Content: What interactions does it have with other systems? How are properties like safety and security handled? The recognition that software contains many different structures has prompted discussion of a number of interesting concepts about software architecture (and software design more generally) ...

Result 3
Source: software_engineering.pdf
BM25 Score: 10.5139
Content: "Goodness" in Architecture [2*, c1.3, 2] [6*, c17] Architecture analysis takes place throughout the process of creating and sustaining an architecture. Architecture evaluation is typically undertaken by third parties at determined milestones as a form of assessment. Given the multi-concern, mu...

Question 9: What is software design?
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Result 1
Source: software_design.pdf
BM25 Score: 9.3940
Content: Design and Information Hiding 15-313: Foundations of Software Engineering Jonathan Aldrich Related Reading: D. L. Parnas. On the Criteria To Be Used in Decomposing Systems into Modules. CACM 15(12):1053-1058, Dec 1972. Some ideas from David Notkin's CSE 503 class
9 September 2008 What makes one desi...

Result 2
Source: software_engineering.pdf
BM25 Score: 8.9865
Content: It is being one- and multi-track-minded simultaneously. [12] What is fundamental about a system varies according to stakeholders' concerns and roles. The software structures, therefore, also vary with stakeholder roles and concerns. (See also topic Design Methods in Software Design KA.) A soft...

22° 08:26 23-08-2026
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Windows PowerShell
Result 3
Source: software_engineering.pdf
BM25 Score: 8.9038
Content: • does a project proposal (a tender) align with a client's business goals? • should certain software functionality be bought or built? • should certain requirements be included in scope or not? • what is the most efficient, cost-effective architecture and design? • what is an optimal load-balancing ...

Question 10: What is the purpose of software engineering?
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Result 1
Source: software_engineering.pdf
BM25 Score: 12.7752
Content: What Are the Objectives of the SWEBOOK Guide? The Guide should not be confused with the body of knowledge itself, which exists in the published literature. The Guide's purpose is to describe the generally accepted portion of the body of knowledge, organize that portion, and provide topical access to...

Result 2
Source: software_engineering.pdf
BM25 Score: 11.9001
Content: Software Engineering Operations Processes and Activities
SOFTWARE ENGINEERING OPERATIONS 6-5 relevant stakeholders for the purpose of specifying, developing, continuously improving, and operating software and system products and services. These processes and activities are the responsibility of o...

Result 3
Source: software_engineering.pdf
BM25 Score: 11.7865
Content: Economics is the science of choice, not the science of money. Engineering economics is the applied microeconomics branch of economics. It asks the fundamental question, "Is it in the best interest of this enterprise to invest its limited resources in this technical endeavor, or would the same ...

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BM25 - 10 QUESTIONS COMPLETED
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22° 08:27 23-08-2026
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Windows PowerShell
Result 3
Source: software_engineering.pdf
Dense Score: 0.4611
Content: A typical unstructured natural language requirements specification is a collection of statements in natural language, such as, "The system shall . . . ."For example, business rules are statements that define or constrain some aspect of the structure or the behavior of the business to be automated. ...

Question 6: What are neural networks?
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Batches: 100%| | 1/1 [00:00<00:00, 59.52it/s]

Result 1
Source: artificial_intelligence.pdf
Dense Score: 0.7597
Content: They are immensely powerful, but also require a lot of computing power and training data. Neural networks only became feasible in the past few years because of the development of more powerful hardware and the rise of big data. Neural networks are inspired by the firing mechanism of biological neuro...

Result 2
Source: artificial_intelligence.pdf
Dense Score: 0.6521
Content: This arrangement is known as a feed-forward network or a fully-connected network. The "firing" of a neuron is calculated by taking a weighted sum over the inputs to the neuron and then applying a nonlinear function to the weighted sum. Other arrangement of neural networks exist for different applica...

Result 3
Source: software_engineering.pdf
Dense Score: 0.5609
Content: Others try to characterize how these dynamic variables evolve over time. The study of neural dynamics can be further subdivided into two separate strands. One tradition, exemplified by the work of Hodgkin and Huxley, focuses on the biophysics of single neurons. The other focuses on the dynamic...

Question 7: What is data processing?
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Batches: 100%| | 1/1 [00:00<00:00, 74.82it/s]

Result 1
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Windows PowerShell
Content: Compound data types also require various ways of traversing data sets to identify specific data items before performing the operation. It is important to ensure that any insertion or deletion of items in a data set or database does not alter the data set or database in a way that violates any policy...

Result 2
Source: software_engineering.pdf
Dense Score: 0.5009
Content: According to [2], a process is a "predetermined course of events defined by its purpose or by its effect, achieved under given conditions."A third definition, following [7], is a "system of activities, which use resources to transform inputs into outputs."A fourth one is a "set of interrelated o...

Result 3
Source: software_engineering.pdf
Dense Score: 0.4914
Content: Software engineers are expected to be aware of the different types of control units, including hardware control units and micro programmable control units (single-level and two-level control stores), along with the benefits and challenges of each. 3. Data Structures and Algorithms [8*, C2] [18*, ...

Question 8: What is software architecture?
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Batches: 100%| | 1/1 [00:00<00:00, 59.58it/s]

Result 1
Source: software_engineering.pdf
Dense Score: 0.8354
Content: 1. Software Architecture Fundamentals [2*, c1] [29*, appendix C] [38*, c2] [41*, c1-3] 1.1. The Senses of "Architecture" [2*, c2] [6*, c6.1] [29*, c6] Software engineering and related disciplines use many senses of "architecture."First, "architecture" often refers to a discipline: the art and science...

Result 2
Source: software_engineering.pdf
Dense Score: 0.7808
Content: Third, "architecture" refers to the outcome of applying architectural design discipline and processes to devise architectures for software systems. Architectures as outcomes are expressed in architecture descriptions. This is discussed in topic Software Architecture Description. The concept...

Result 3
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Windows PowerShell
Source: software_engineering.pdf
Dense Score: 0.7763
Content: The discipline involves concepts, principles, processes and methods the community has discovered and adopted. Second, "architecture" refers to the various processes through which that discipline is realized. Software architecture is also considered part of Software Design; generally considered...

Question 9: What is software design?
-----
Batches: 100%| 1/1 [00:00<00:00, 52.60it/s]

Result 1
Source: software_engineering.pdf
Dense Score: 0.7176
Content: Design is used in distinct but closely related ways to refer to (1) the discipline ("use of scientific principles, technical information, and imagination in the definition of a software system to perform [prespecified] functions with maximum economy and efficiency") [11]; (2) the processes for pe...

Result 2
Source: software_engineering.pdf
Dense Score: 0.7032
Content: 3-2 SWEBOOK @ GUIDE V4.0a 1. Software Design Fundamentals [3*][4*] The concepts, notions and terminology introduced here form a basis for understanding the role and scope of software design. 1.1. Design Thinking [3*, c1-3] [4*, c1-2] Design is all around us, in the things and organizations that ...

Result 3
Source: software_engineering.pdf
Dense Score: 0.6848
Content: Ross, Goodenough and Irvine offer an elaboration of design thinking appropriate to software: This process consists of five basic steps: (1) crystallize a purpose or objective; (2) formulate a concept for how the purpose can be achieved; (3) devise a mechanism that implements the conceptual structure...

Question 10: What is the purpose of software engineering?
-----
Batches: 100%| 1/1 [00:00<00:00, 56.98it/s]

Result 1
Source: software_engineering.pdf
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Windows PowerShell
Dense Score: 0.6472
Content: 4. Software Engineering Methods Software engineering methods provide an organized and systematic approach to developing software for a target computer. There are numerous methods from which to choose, and the software engineer needs to choose an appropriate method or methods for the software development...

Result 2
Source: software_engineering.pdf
Dense Score: 0.6456
Content: 1. Software Engineering Process Fundamentals 1.1 Introduction [1,c5],[13] Software engineering processes involve work activities software engineers conduct to build and operate software. When the discipline of software engineering emerged, scientists, engineers and technicians had to look at existing...

Result 3
Source: software_engineering.pdf
Dense Score: 0.6357
Content: In engineering, science and practice are applied to generate potential solutions to the real-world problem, and engineering economics is used to identify the most cost-effective one. In the same way that it would not make sense to send a chemist to solve a chemical engineering problem, it does not...

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DENSE RETRIEVAL - 10 QUESTIONS COMPLETED
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BM25 VS DENSE RETRIEVAL COMPARISON
=====

Evaluation method:
Precision@3 = Number of relevant documents in Top-3
              -----
              3

Because this PDF collection has no predefined relevance labels, relevance was evaluated manually.
```

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Windows PowerShell

Q  BM25 Relevant      BM25 P@3   Dense Relevant   Dense P@3   Winner
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1  3/3              1.00      2/3            0.67       BM25
2  3/3              1.00      3/3            1.00       Tie
3  2/3              0.67      3/3            1.00       Dense
4  1/3              0.33      3/3            1.00       Dense
5  2/3              0.67      1/3            0.33       BM25
6  1/3              0.33      2/3            0.67       Dense
7  1/3              0.33      1/3            0.33       Tie
8  3/3              1.00      3/3            1.00       Tie
9  3/3              1.00      3/3            1.00       Tie
10 2/3              0.67      3/3            1.00       Dense

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OVERALL RETRIEVAL QUALITY
=====

BM25 Mean Precision@3: 0.70
Dense Mean Precision@3: 0.80

Overall Winner: Dense Retrieval

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W7D2 - HAYSTACK RETRIEVAL TASK STATUS
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Completed:
[✓] Found and indexed 5 PDF documents
[✓] Converted PDF documents
[✓] Cleaned documents
[✓] Split documents into chunks
[✓] Created BM25 document store
[✓] Created BM25 retriever
```

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Windows PowerShell

Dense Mean Precision@3: 0.80

Overall Winner: Dense Retrieval

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W7D2 - HAYSTACK RETRIEVAL TASK STATUS
=====

Completed:
[✓] Found and indexed 5 PDF documents
[✓] Converted PDF documents
[✓] Cleaned documents
[✓] Split documents into chunks
[✓] Created BM25 document store
[✓] Created BM25 retriever
[✓] Created BM25 Haystack pipeline
[✓] Ran 10 questions using BM25
[✓] Created Sentence Transformer document embeddings
[✓] Created Sentence Transformer query embeddings
[✓] Created Dense document store
[✓] Created Dense retriever
[✓] Created Dense Haystack pipeline
[✓] Ran the same 10 questions using Dense Retrieval
[✓] Compared BM25 and Dense Retrieval
[✓] Completed manual Precision@3 evaluation

BM25 Mean Precision@3: 0.70
Dense Mean Precision@3: 0.80
Overall Winner: Dense Retrieval

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W7D2 - HAYSTACK RETRIEVAL IMPLEMENTATION COMPLETED
=====
PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W7D2_Haystack_Retrieval>
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